

Civil Engineering

Trigonometry and Complex Numbers

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$c^2 = a^2 + b^2 - 2ab \cos C$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$z = a + bi$$

Section B

5. Compute the height of a structure from angle of elevation. **(10 marks)**
6. Solve a triangular road reserve survey problem. **(10 marks)**
7. Convert a complex number to polar form. **(10 marks)**
8. Create one realistic civil engineering problem based on this topic and solve it fully. **(20 marks)**